```r
library(tidyr)
library(dplyr)
library(ggplot2)
```

1 GLM for Binary Data

This uses the snoring levels as predictor to occurrence of heart attack. Refer to pg 118, Agresti.

```r
scoring <- c(0,2,4,5)

snoring<-rep(scoring,c(1379, 638, 213, 254))
y<-rep(rep(c(1,0),4),c(24,1355,35,603,21,192,30,224))
snoredata <- data.frame(y,snoring)
# snoredata$y <- factor(snoredata$y, levels = c(1,0), ordered = TRUE)
# snoredata$snoring <- factor(snoredata$snoring,
#                              levels = c(0,2,4,5), ordered = TRUE)

head(snoredata)
```

```
|    y snoring
|  1 1       0
|  2 1       0
|  3 1       0
|  4 1       0
|  5 1       0
|  6 1       0
```

Here is the contingency table:

```r
groupsnore <- data.frame(count=c(1379,638,213,254),
                         snore=scoring,
                         heart.disease = c(24,35,21,30))

groupsnore
```

```
|    count snore heart.disease
|  1  1379     0            24
|  2   638     2            35
|  3   213     4            21
|  4   254     5            30
```

```r
contingency_tab <- snoredata %>%
  group_by(snoring) %>%
  count(y) %>%
  pivot_wider(names_from = y, values_from = n) %>%
  mutate(totals = `1` + `0`)

contingency_tab
```

```
|  # A tibble: 4 x 4
|  # Groups:   snoring [4]
|    snoring   `0`   `1` totals
|      <dbl> <int> <int>  <int>
|  1       0  1355    24   1379
|  2       2   603    35    638
|  3       4   192    21    213
|  4       5   224    30    254
```

1.1 Linear Probability Model

For binary response variable, the regression model

$$\pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_\rho x_\rho$$

is called a *linear probability model*. With independent observations, it is a GLM with binomial random component and identity link function.

(-): Probabilities fall between 0 and 1, but linear functions take values over the entire real line. The model can have $\pi(\mathbf{x}) > 1$ for some $\mathbf{x}$ values.

(+): its simple interpretation: β_j *is the change in* $\pi(\mathbf{x})$ for a one-unit increase in x_j

Since $var(Y) = \pi\left(\mathbf{x}\left[1 - \pi(\mathbf{x})\right]\right)$, the variance depends on $\mathbf{x}$ through its influence on $\pi(\mathbf{x})$. The constant variance condition that makes OLS estimators optimizal is not satisfied, so the ML estimator is more efficient that least squares.

[Google AI] While glm with `family=binomial` is typically used for logistic regression (which uses a "logit" link), this uses `link="identity"`. This forces the model to treat the relationship between the predictors and the probability of the outcome as a straight line rather than a curve.

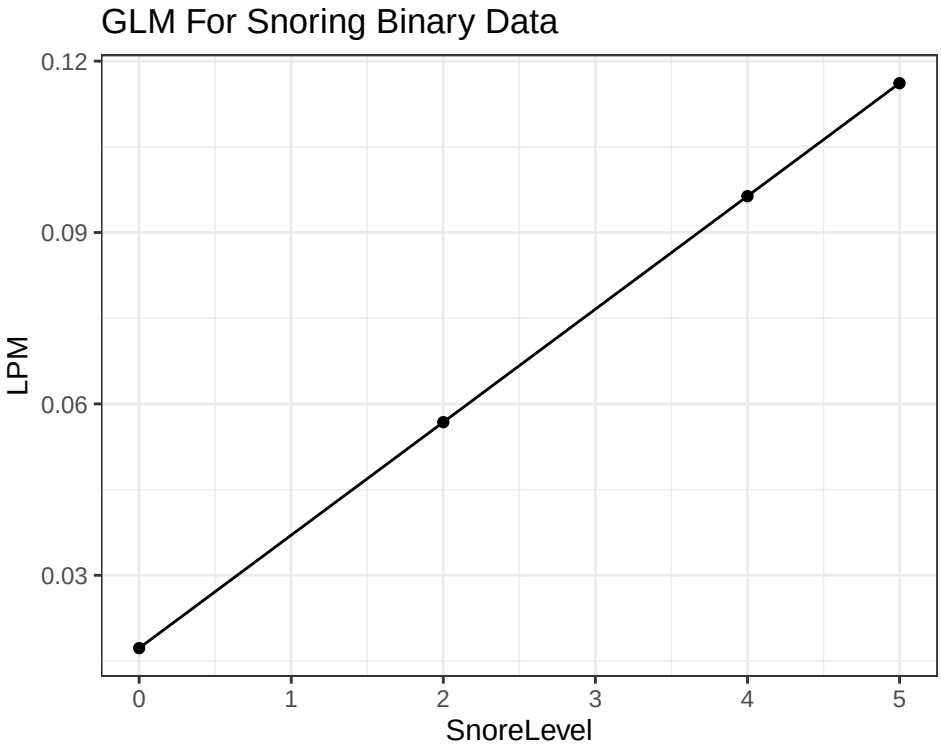
```r
fit <- glm(y~snoring,data=snoredata,family=binomial(link="identity"))
fit
```

```
|
|  Call:  glm(formula = y ~ snoring, family = binomial(link = "identity"),
|      data = snoredata)
|
|  Coefficients:
|  (Intercept)      snoring
|       0.0172       0.0198
|
|  Degrees of Freedom: 2483 Total (i.e. Null);  2482 Residual
|  Null Deviance:       901
|  Residual Deviance: 835   AIC: 839
```

For nonsnorers (`x=0`), the estimated proportion of subjects having heart disease is 0.01725. The estimated probability of heart disease is about 0.01725 for nonsnorers and it increases 2 × 0.01978=0.03956 for occasional snorers (`snoring = 2`), 4 × 0.01978=0.07911 for those who snore nearly every night (`snoring = 4`), and 5 × 0.01978=0.09889 for those who always snore (`snoring = 5`)

```r
new_obs <- data.frame(snoring = c(0, 2, 4, 5))
LPM_predictions <- predict(fit, newdata = new_obs, type = "response")
```

```r
data.frame(`SnoreLevel` = new_obs$snoring, LPM = LPM_predictions) %>%
  ggplot(aes(x = `SnoreLevel`, y = LPM)) +
  geom_point() +
  geom_line() +
  labs(title = "GLM For Snoring Binary Data") +
  theme_bw()
```



[Agresti] ML estimates and se are the same if we use a data file of 2484 binary observations (as shown above) or if we enter the four binomial totals of "yes" and "no" responses using `contingency_tab` (as shown below). Note: Set the first column as the success occurrence so that we get the same values as above.

```r
glm(cbind(contingency_tab$`1`, contingency_tab$`0`) ~ contingency_tab$snoring,
    family=binomial(link="identity"))
```

```
|
|  Call:  glm(formula = cbind(contingency_tab$‘1‘, contingency_tab$‘0‘) ~
|      contingency_tab$snoring, family = binomial(link = "identity"))
|
|  Coefficients:
|              (Intercept)  contingency_tab$snoring
|                   0.0172                   0.0198
|
|  Degrees of Freedom: 3 Total (i.e. Null);  2 Residual
|  Null Deviance:       65.9
|  Residual Deviance: 0.0692    AIC: 24.3
```

See that it reverses when columns reverse. It just differs in interpretation.

```r
altfit <- glm(
  cbind(contingency_tab$`0`, contingency_tab$`1`) ~ contingency_tab$snoring,
  family=binomial(link="identity"))
altfit
```

```
|
|  Call:  glm(formula = cbind(contingency_tab$`0`, contingency_tab$`1`) ~
|      contingency_tab$snoring, family = binomial(link = "identity"))
|
|  Coefficients:
|              (Intercept)  contingency_tab$snoring
|                   0.9828                  -0.0198
|
|  Degrees of Freedom: 3 Total (i.e. Null);  2 Residual
|  Null Deviance:       65.9
|  Residual Deviance: 0.0692    AIC: 24.3
```

The estimated probability of not having heart disease is about 0.98275 for nonsnorers and it decreases $-1\times$ 2 $\times$ -0.01978=0.03956 for occasional snorers (`snoring = 2`), $-1\times$ 4 $\times$ -0.01978=0.07911 for those who snore nearly every night (`snoring = 4`), and $-1\times$ 5 $\times$ -0.01978=0.09889 for those who always snore (`snoring = 5`)

Notes:

- you get completely different model when you treat `snoring` as categorical rather than continuous. It depends how you want to model. Treating it as continuous is parsimonious.

1.2 Logistic regression

Usually binary data result from a nonlinear relationship between $\pi(x)$ and x. A fixed change in x often has less impact when $\pi(x)$ is near 0 or 1 than when $\pi(x)$ is near 0.50.

In practice, nonlinear relationships between $\pi(x)$ and x are often monotonic, with $\pi(x)$ increasing continuously or $\pi(x)$ decreasing continuously as x increases. The most important curve with this shape has the model formula,

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}.$$

This is a logistic regression model. As x increases, $\pi(x)$ increases when $\beta > 0$ and decreases when $\beta < 0$. Logistic regression is a GLM, the link is the log odds transformation — the logit. Logistic regression models are GLMs with binomial random component and logit link function.

$$\begin{aligned}
\pi(x) &= \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \\
\pi(x) + \pi(x)\exp(\alpha + \beta x) &= \exp(\alpha + \beta x) \\
\pi(x) &= \exp(\alpha + \beta x) - \pi(x)\exp(\alpha + \beta x) \\
\frac{\pi(x)}{1 - \pi(x)} &= \exp(\alpha + \beta x) \\
\log \frac{\pi(x)}{1 - \pi(x)} &= \alpha + \beta x
\end{aligned}$$

Whereas $\pi(x)$ must fall in the $(0, 1)$ range, the logit can be any real number. The real numbers are also the range for linear predictors that form the systematic component of a GLM. So, this model does not have the structural problem that LPM has.

```
groupsnore$snore
```

```
|  [1] 0 2 4 5
```

```
groupsnore$count
```

```
|  [1] 1379  638  213  254
```

```
groupsnore$heart.disease
```

```
|  [1] 24 35 21 30

|  function (x, ..., wt = NULL, sort = FALSE, name = NULL)
|  {
|      UseMethod("count")
|  }
|  <bytecode: 0x000001af3307bba0>
|  <environment: namespace:dplyr>
```

```
fit2 <-
fit2
```

```
logitfit <- glm(
  cbind(contingency_tab$`1`, contingency_tab$`0`) ~ contingency_tab$snoring,
  family=binomial(link="logit"))
logitfit
```

```
|
|  Call:  glm(formula = cbind(contingency_tab$‘1‘, contingency_tab$‘0‘) ~
|      contingency_tab$snoring, family = binomial(link = "logit"))
|
|  Coefficients:
|              (Intercept)  contingency_tab$snoring
|                   -3.866                    0.397
|
|  Degrees of Freedom: 3 Total (i.e. Null);  2 Residual
|  Null Deviance:       65.9
|  Residual Deviance: 2.81  AIC: 27.1
```

```
# same
# glm(contingency_tab$`1`/contingency_tab$totals~contingency_tab$snoring,
# family=binomial(link="logit"),weights=contingency_tab$totals)
```